Data-Level vs. Algorithm-Level Solutions for Imbalanced Classification: A Comparative Study Using Multiple Machine Learning Models

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Abstract

Imbalanced datasets present a significant challenge in supervised machine learning because one class often contains substantially more examples than another. This project compares data-level and algorithm-level techniques for handling class imbalance using the Credit Card Fraud Detection and Breast Cancer Wisconsin datasets. Logistic Regression and Random Forest were evaluated using Baseline, Random Oversampling, Random Undersampling, SMOTE, and Class Weighting. In addition, Balanced Random Forest and a Neural Network trained with Focal Loss were included as specialized algorithm-level approaches for handling class imbalance. Performance was evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, confusion matrices, ROC curves, class distribution plots, and F1 score comparisons. The results demonstrate that no single imbalance-handling strategy consistently produced the best performance across both datasets; instead, performance depended on the dataset, learning algorithm, and evaluation metric.

1. Introduction

Many real-world machine learning applications require models to identify rare but important events, such as fraudulent financial transactions, disease diagnosis, spam detection, and cybersecurity threats. In these applications, the events of greatest interest often represent only a small portion of the available data, creating an imbalanced classification problem in which machine learning models tend to favor the majority class while struggling to correctly identify minority-class examples. Because overall accuracy alone can provide a misleading assessment of performance on imbalanced datasets, additional evaluation metrics such as Precision, Recall, F1 Score, and ROC-AUC are commonly used to evaluate model effectiveness. To address this challenge, researchers have developed both data-level techniques, which modify the training data, and algorithm-level techniques, which adjust how learning algorithms handle class imbalance. This project compares several of these approaches using the Credit Card Fraud Detection and Breast Cancer Wisconsin datasets to investigate which imbalance-handling strategies produce the strongest classification performance across different levels of class imbalance.

2. Research Question

The purpose of this study is to investigate how different data-level and algorithm-level imbalance-handling techniques affect classification performance on datasets with varying levels of class imbalance. Specifically, this study seeks to answer the following research question:

Which data-level and algorithm-level imbalance-handling strategy produces the strongest classification performance across datasets with different levels of class imbalance?

To address this research question, Logistic Regression, Random Forest, Balanced Random Forest, and a Neural Network trained with Focal Loss were evaluated using multiple imbalance-handling techniques. Model performance was assessed using Accuracy, Precision, Recall, F1 Score, ROC-AUC, confusion matrices, and ROC curves.

3. Methodology

This project evaluated four supervised machine learning classifiers: Logistic Regression, Random Forest, Balanced Random Forest, and a Neural Network trained with Focal Loss.

- Baseline (no imbalance handling)
- Random Oversampling
- Random Undersampling
- Synthetic Minority Oversampling Technique (SMOTE)
- Class Weighting
- Balanced Random Forest
- Focal Loss

The Credit Card Fraud Detection dataset contains more than 280,000 transactions, with fraudulent transactions representing less than one percent of the data. To reduce computation time, the majority class in the training set was randomly limited to 25,000 legitimate transactions while preserving all fraudulent transactions. The 25,000 legitimate transactions were selected randomly using a fixed random seed (random_state = 42) to ensure reproducibility. The Breast Cancer Wisconsin dataset was evaluated using its original class distribution.

Both datasets were divided into training and testing sets using a stratified train-test split. Logistic Regression used standardized input features before training, while the remaining models followed the same evaluation pipeline. Performance was evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, confusion matrices, ROC curves, class distribution plots, and F1 score comparison charts.

4. Results

The effectiveness of each imbalance-handling strategy depended on both the dataset and the learning algorithm.
For the Credit Card Fraud dataset, Random Forest generally produced the strongest overall performance. Random Forest with SMOTE achieved the highest ROC-AUC, while the baseline Random Forest model achieved the highest F1 Score. Balanced Random Forest and the Neural Network trained with Focal Loss also produced competitive results.

For the Breast Cancer Wisconsin dataset, nearly every model achieved excellent classification performance because the classes were more easily separable and less severely imbalanced. Logistic Regression produced the strongest overall performance, while Balanced Random Forest and the Neural Network with Focal Loss remained competitive. The ROC curves and F1 score comparisons further demonstrated that the model with the highest ROC-AUC was not always the same model with the highest F1 Score.

5. Discussion

The experiments demonstrate that no single imbalance-handling strategy consistently outperformed all others. Performance depended on the characteristics of the dataset, the learning algorithm, and the evaluation metric. Random Forest-based approaches generally performed best on the highly imbalanced fraud dataset, while Logistic Regression remained highly competitive on the breast cancer dataset. These findings also demonstrate that Accuracy alone is insufficient when evaluating imbalanced datasets. Precision, Recall, F1 Score, and ROC-AUC together provide a more comprehensive assessment of model performance, particularly for minority-class prediction..

6. Conclusion

This project compared multiple data-level and algorithm-level approaches for handling imbalanced classification problems using Logistic Regression, Random Forest, Balanced Random Forest, and a Neural Network trained with Focal Loss. The results showed that the most effective strategy depended on the dataset and evaluation metric rather than a single universally best method. The inclusion of Balanced Random Forest and a Neural Network trained with Focal Loss broadened the comparison by incorporating additional algorithm-level approaches specifically designed for imbalanced learning. Although these methods produced competitive results, no single technique consistently outperformed the others across every dataset and evaluation metric.

7. GitHub Repository

https://github.com/Natnail101/imbalanced-classification-comparison